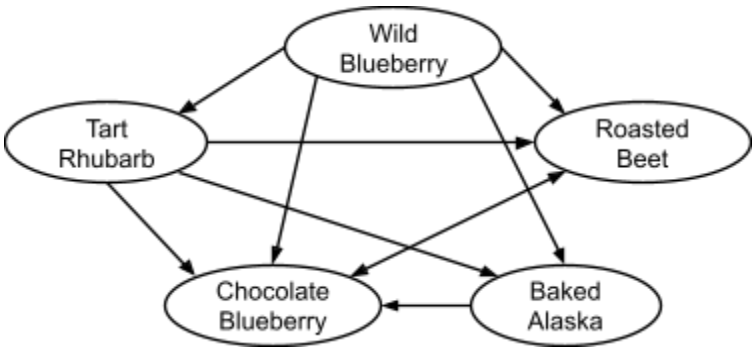
American Computer Science League

2025-2026 ● Contest 4: Short Problems ● Intermediate Division

1. Graph Theory

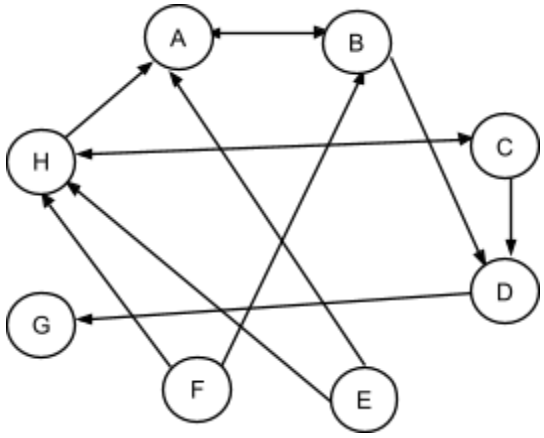
Sally goes to Wild Scoops to buy a 3-scoop ice cream cone. She needs to choose which scoop is on the bottom, middle, and top. The rules of the shop say that the scooper can only be used in a certain order. For example, from the graph below, if you scoop Baked Alaska, then only Chocolate Blueberry can go on top of it. Which of the following is NOT possible from bottom to top?



A. Tart Rhubarb, Chocolate Blueberry, Roasted Beet
B. Baked Alaska, Chocolate Blueberry, Roasted Beet
C. Wild Blueberry, Tart Rhubarb, Baked Alaska
D. Wild Blueberry, Baked Alaska, Chocolate Blueberry
E. Roasted Beet, Tart Rhubarb, Baked Alaska

2. Graph Theory

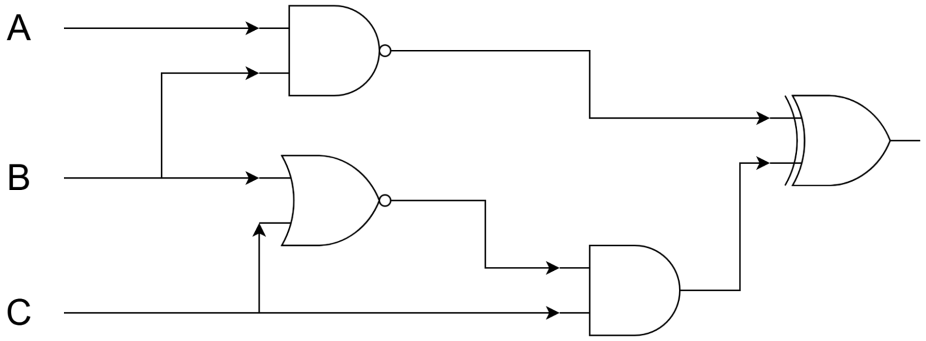
How many vertices do not appear in any cycles of this directed graph?



A. 3
B. 4
C. 5
D. 6
E. 7

3. Digital Electronics

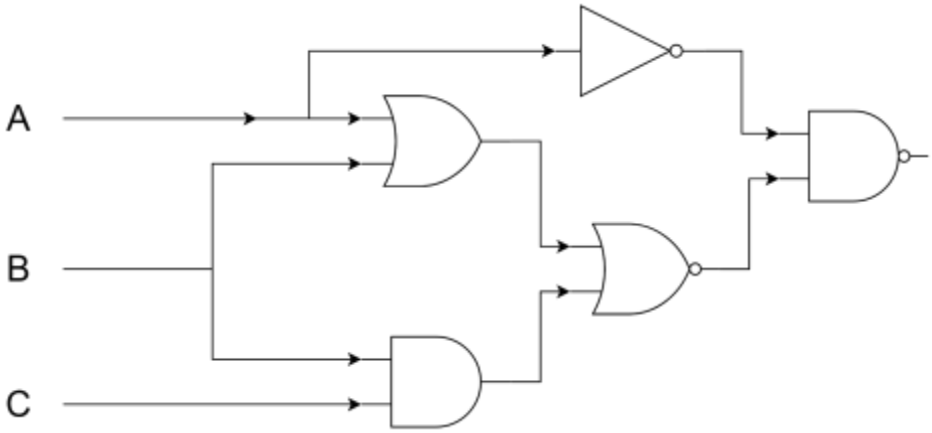
How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. 8

4. Digital Electronics

Simplify the Boolean algebra expression represented by the following digital circuit:



A. $\overline{A}$
B. ABC
C. $A + BC$
D. $A + B$
E. BC

5. Assembly Language

What is the output when the following program is executed?

```
A       DC      7
B       DC      8
        LOAD    B
        SUB     =10
        MULT    =-1
        STORE   C
        LOAD    B
        MULT    C
        STORE   D
        LOAD    A
        MULT    B
        MULT    =100
        ADD     D
        STORE   ANS
        PRINT   ANS
        END
```

A. 72
B. 1224
C. 3024
D. 4221
E. 5616

☐

6. Assembly Language

What is the output when the following program is executed with these data values: 24, 11, 22, 35, 13, 26, 7, 0, -3?

```
ANS     DC      0
        READ    P
TOP     READ    N
        LOAD    N
        BE      STOP
        LOAD    P
        SUB     N
        BL      CONT
        LOAD    ANS
        ADD     =1
        STORE   ANS
CONT    LOAD    N
        STORE   P
        BU      TOP
STOP    PRINT   ANS
        END
```

A. 0
B. 1
C. 2
D. 3
E. 4